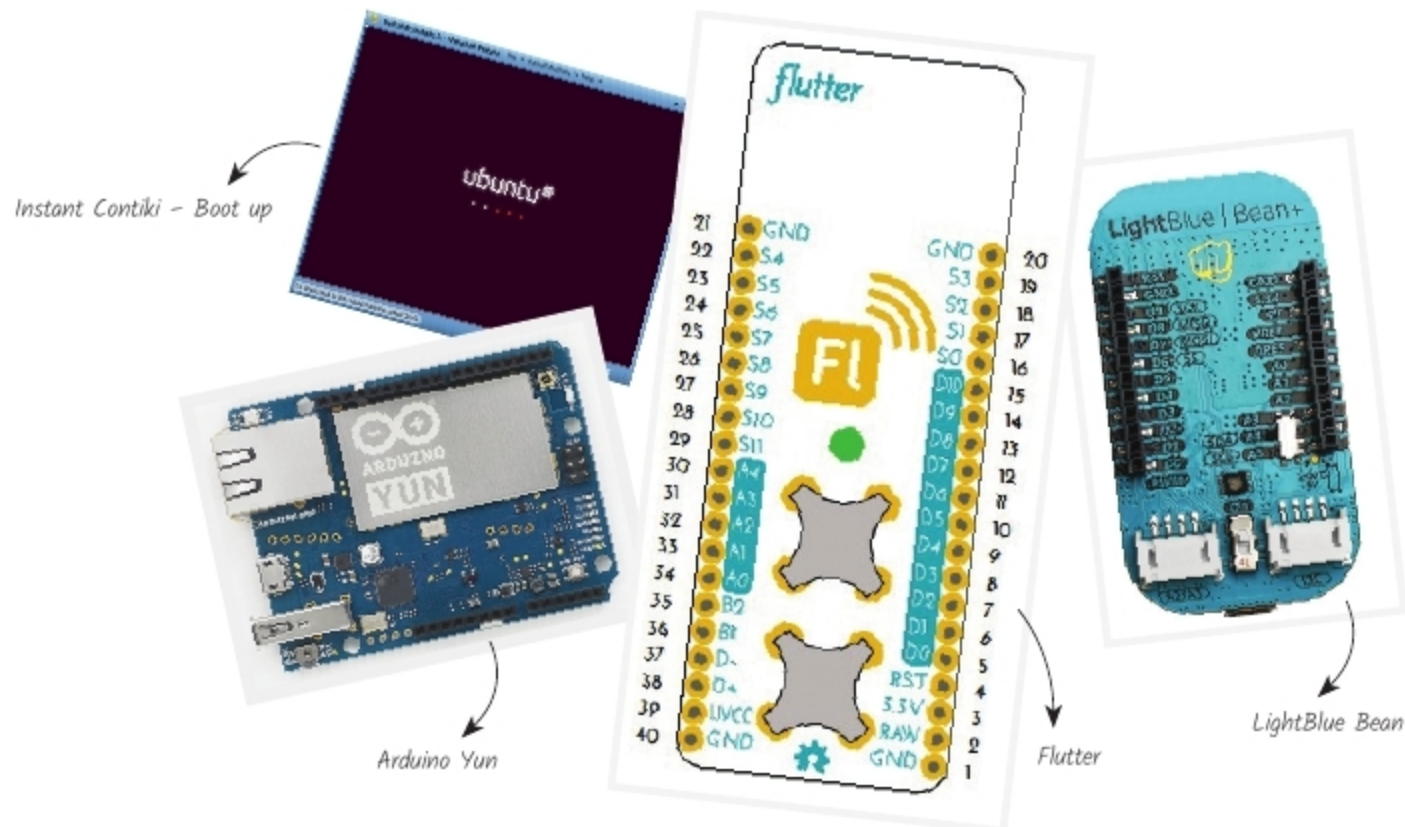


A Quick Look at Open Source Tools and Services for IoT

IoT is gaining ground in our lives at a very rapid rate. This makes the study of IoT and its related aspects very important. This article presents open source IoT tools and services in a nutshell.



The term ‘open source’ is primarily related to software in which the source code is freely accessible to users other than the developer—for examination, use and modification or expansion.

The growth of the Internet of Things (IoT) is based on the continuous developments in the fields of microelectronics, IT and communications, and it is evident this trend will continue. Openness has been the most noted driver of the IoT revolution. It has helped create many interesting applications, ranging from automated kettles to data analysers.

As IoT comprises connected objects like smart appliances, machinery, wheels, etc, the user experience with it relies on the physical structure, its specifications and the way it’s been applied.

What drives open source

There are three main elements behind the growth of the open source movement.

- **Beneficiaries:** They wish to use a wide category of consumer technologies and not just those from a particular vendor.
- **IoT service providers:** They have developed various technologies, paving the way to versatile compatibility.
- **Developers:** They develop openly accessible code supporting a range of devices, platforms and technologies, rather than being locked with just one vendor.

What follows is a bird’s eye view of the various open source tools and services that will help in the development of IoT.

Open source IoT development tools

Arduino is an open source prototyping platform with both software and hardware offering ease of use. It is also interactive, with a simple programming language and an IDE.

Kinoma is a prototyping hardware platform from Marvell Semiconductor, which controls over three open source projects, namely, Kinoma Create, a DIY construction kit; Kinoma Studio, a development environment; and Kinoma Connect, a free app for Android and iOS, connecting IoT devices and mobile devices.

The **Eclipse** project sponsors numerous projects on IoT, including application services and frameworks, open source implementation of IoT protocols, and Lua development tools.

M2M Labs is another open source framework for building M2M applications such as smart grids, fleet management, remote monitoring, etc. Its capabilities include flexible device modelling, device configuration, device and application communication, data validation and normalisation, prolonged data storage and recovery. It is based on the Apache Cassandra NoSQL database.

Open source hardware

Arduino Yun combines the microcontroller and Linux for the Arduino board. It comprises the ATmega32u4 (which supports Arduino) and the Atheros AR9331 (which runs on Linux) processors. Its built-in features are Wi-Fi, Ethernet, a USB port, microSD card, reset buttons, etc.

BeagleBoard offers credit card-sized boards that run Android and Linux. Since its power requirements are very low,